

Computational Sociology

Introduction

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Plan

- ▶ Part I
 - ▶ Introductions
 - ▶ A brief introduction to computational sociology
 - ▶ Course outline
 - ▶ AI use
- ▶ Part II
 - ▶ File management
 - ▶ R, RStudio, and RMarkdown
 - ▶ GitHub

Computational Sociology

Our World
in Data

in Data

e and Max Roser

e and Max Roser

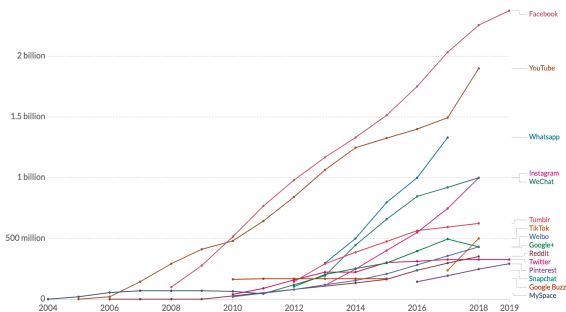
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Number of people using social media platforms, 2004 to 2019

Estimates correspond to monthly active users (MAUs). Facebook, for example, measures MAUs as users that have logged in during the past 30 days. See source for more details.

Our World
in Data

+ Add



<https://ourworldindata.org/grapher/users-by-social-media-platform>

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Computational Social Science

The capacity to collect and analyze massive amounts of data has unambiguously transformed such fields as biology and physics. The emergence of such a data-driven “computational social science” has been much slower, largely spearheaded by a few intrepid computer scientists, physicists, and social scientists. If one were to look at the leading disciplinary journals in economics, sociology, and political science, there would be minimal evidence of an emerging computational social science engaged in quantitative modeling of these new kinds of digital traces. However, computational social science is occurring, and on a large scale, in places like Google, Yahoo, and the National Security Agency. Computational social science could easily become the almost exclusive domain of private companies and government agencies. Alternatively, there might emerge a “Dead Sea Scrolls” model, with a privileged set of academic researchers sitting on private data from which they produce papers that cannot be critiqued or replicated. Neither scenario will serve the long-term public interest in the accumulation, verification, and dissemination of knowledge.

Lazer et al. 2009

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Digital traces and big data

[F]ust as the invention of the telescope revolutionized the study of the heavens, so too by rendering the unmeasurable measurable, the technological revolution in mobile, Web, and Internet communications has the potential to revolutionize our understanding of ourselves and how we interact [T]hree hundred years after Alexander Pope argued that the proper study of mankind should lie not in the heavens but in ourselves, we have finally found our telescope. Let the revolution begin.

—Duncan Watts (2011, p. 266)

Quoted in Golder and Macy 2014

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Digital traces and big data

“[S]ocial scientists—and cultural sociologists in particular—have largely ignored the promise of so-called “big data.” Instead, cultural sociologists have left this wellspring of information about the arguments, worldviews, or values of hundreds of millions of people from Internet sites and other digitized texts to computer scientists who possess the technological expertise to extract and manage such data but lack the theoretical direction to interpret their meaning.”

Bail 2014

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Example: Measuring diurnal rhythms using Twitter

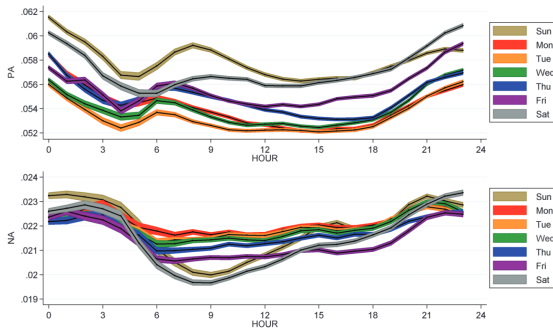


Fig. 1. Hourly changes in individual affect broken down by day of the week (top, PA; bottom, NA). Each series shows mean affect (black lines) and 95% confidence interval (colored regions).

(PA = Positive Affect; NA = Negative Affect. Golder and Macy 2011)

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Readymade data



Readymade



Custommade

Salganik 2017 https://www.bitbybitbook.com/figures/chapter1/bitbybit1-2_readymade-custommade.png

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Is it quantitative or qualitative?

- ▶ Early work in CSS largely used quantitative methods like social network analysis and simulation, as well as conventional statistical techniques
- ▶ New approaches to text that complement and extend qualitative methods
 - ▶ Topic modeling as content analysis (e.g., DiMaggio, Nag, and Blei 2013; Mohr and Bogdanov 2013)
 - ▶ Computational ethnography (Abramson et al. 2018, 2026)
 - ▶ Qualitative research with LLM chatbots (Ibrahim and Voyer 2026)
- ▶ Hybrid, mixed-methods approaches
 - ▶ Computational grounded theory (Nelson 2020)

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The expansion of a field

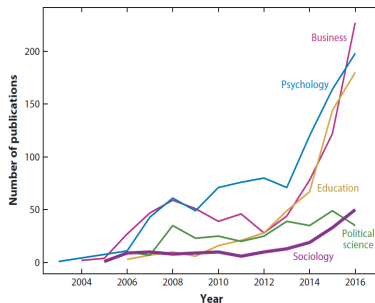


Figure 1

Number of computational social science publications by year—2003–2016—across four scholarly disciplines.

Edelmann et al. 2020

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Toward a definition

1. Use of readymade, multimodal data (image, text, video)
2. New modes of data collection (web-scraping, APIs, online surveys and experiments)
3. Application of methods developed in computer science (topic modeling, word embeddings, deep learning, generative AI)

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Generative AI and transformation of the field

- ▶ Generative AI as a General Purpose Technology
 - ▶ AI as method
 - ▶ AI as tool
 - ▶ AI as object
- ▶ Is all sociology computational now?

Course outline

Goals

- ▶ By the end of this course, students will be able to
 - ▶ Understand and implement fundamental data structures and programming concepts
 - ▶ Collect observational data from online sources and build familiarity with online surveys and experiments
 - ▶ Apply a range of computational techniques to social science research questions, and understand how these techniques can complement conventional quantitative and qualitative approaches
 - ▶ Critically evaluate the use of new data sources and computational models, including methodological and ethical challenges

Course outline

Structure

1. Programming and data structures (Weeks 1-2)
2. Data collection (Weeks 3-4)
3. Natural language processing (Weeks 5-7)
4. Machine learning (Weeks 8-9)
5. Large language models and multimodal data (Weeks 10-12)

Course outline

Assessment

- ▶ Homework assignments: 30%
- ▶ Presentations: 20%
- ▶ Final project: 50%

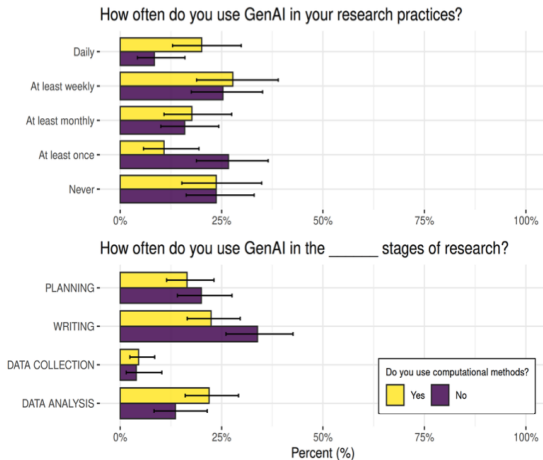
Generative AI

- ▶ Large language models are trained on large amounts of text data as well as code from GitHub
 - ▶ This means they can “understand” R and can write and comment on code
 - ▶ And these models have “knowledge” about most of the topics covered during the class
 - ▶ Writing, executing, and debugging code

Generative AI

- ▶ Generative AI tools are now a routine part of computational research, whether as coding assistants, as agents that carry out multi-step analytic tasks, or as measurement instruments
- ▶ I expect you to develop an informed judgment about them
 - ▶ That judgment comes from using these tools enough to learn where they help and where they fail
 - ▶ And from studying them as objects of inquiry

AI adoption in sociology



Alvero et al. 2026

Emerging norms about AI use



Stuhler et al. 2026 (working paper)

Generative AI

Applications

- ▶ Generative AI as a *research method*
 - ▶ Later in the semester, we will discuss these technologies in more detail and how they can advance computational social science
- ▶ Generative AI as a *pedagogical tool*
 - ▶ These tools can help you learn computational methods, but they also pose challenges to academic integrity

Generative AI as a pedagogical tool

Benefits and risks

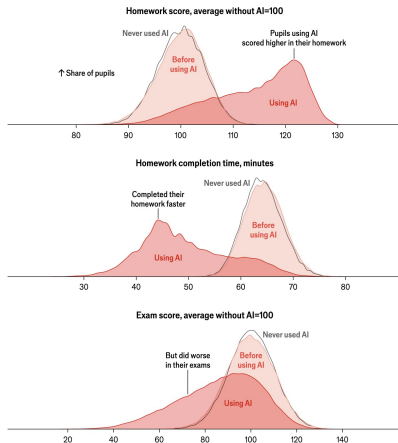
▶ *Benefits*

- ▶ Bespoke problem solving
- ▶ Efficient for coding and debugging
- ▶ Deep knowledge base, search augmentation

▶ *Risks*

- ▶ Undermines learning
- ▶ Replaces thinking; cognitive atrophy
- ▶ Insufficient contextual grounding

Negative impacts on learning



Source: "The generative AI learning penalty: evidence from Chinese secondary education", by D. Strömberg et al., preprint June 2026

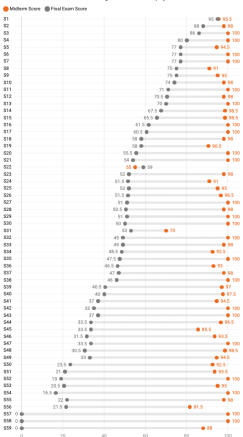
*26,811 pupils aged 12-18

<https://www.economist.com/graphic-detail/2026/08/18/does-ai-stop-children-from-learning>

Negative impacts on learning

ECON 1170 Midterm & Final Scores

After Senemo made the final in-person, 18 students dropped the class and nine students did not take the exam. The scores of the remaining 59 students are displayed here.



<https://www.insidehighered.com/news/faculty/learning-assessment/2026/07/08/brown-professor-suspects-most-his-class-used-ai-cheat>

AI use in this course

Three types of permitted use

▶ **No AI**

- ▶ Exercises designed to build fundamentals
- ▶ Required readings

▶ **AI-assisted work**

- ▶ Generate or debug code
 - ▶ You must understand all code you submit
- ▶ *Never* permitted for writing

▶ **AI as a method**

- ▶ Using AI as a measurement tool
- ▶ Justify and validate

AI use in this course

Responsibility

- ▶ You are responsible for whatever you submit
 - ▶ If a model makes an error and you miss it, it is your error
- ▶ Verify, test, and explain anything you hand in
- ▶ *Include a short statement describing how you used AI tools in all assignments*
- ▶ Ask me if you are unsure whether a particular use is acceptable

Exercise

Problem solving in R

- ▶ Download and inspect this code in RStudio (paste it into a new R file)
 - ▶ <https://tinyurl.com/code-to-fix>
- ▶ Task: Identify and fix the errors

Exercise

Problem solving in R

- ▶ Three groups:
 1. Fix it without using the internet
 2. Fix it by searching StackOverflow as needed
 3. Fix it by using AI as needed (ChatGPT, Gemini, etc.)

Break

File management



Source: The Verge, 2021.

File management

- ▶ Keeping track of course materials
- ▶ Working on homework assignments
- ▶ Organizing final projects
- ▶ Reproducibility

Organizing your files

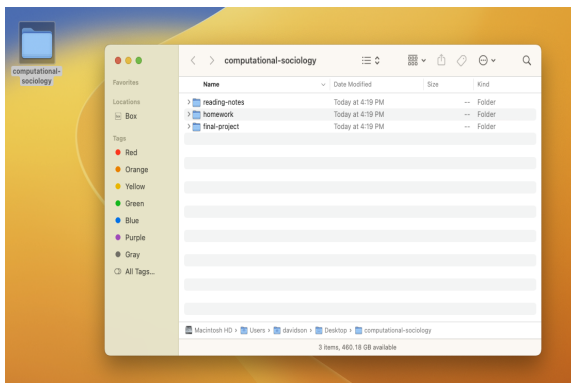
- ▶ Make a directory to contain all materials for this class
- ▶ Store this somewhere practical
 - ▶ e.g., `/Users/me/Documents/computational-sociology`,
`/Users/me/Desktop/Classes/computational-sociology`
 - ▶ Do not leave files in your Downloads directory!

Organizing your files

- ▶ Within this directory, create separate directories for other files, like readings and homework assignments
- ▶ Do not use spaces in directory names! Use dashes or underscores instead

Organizing your files

It might look something like this once you have added other sub-directories.



Why R?



<https://kieranhealy.org/blog/archives/2019/02/07/statswars/>

Why R?

- ▶ Free and open-source
- ▶ A statistical programming language
- ▶ Active developer community
 - ▶ Many cutting-edge approaches now implemented in R before Stata

RStudio

Overview

- ▶ RStudio is an Integrated Development Environment (IDE) for programming in R
 - ▶ Run code
 - ▶ View data, objects, plots
 - ▶ Create output like papers or slides
 - ▶ Integrations including GitHub and a Python interpreter

RMarkdown

Overview

- ▶ RMarkdown is an interactive coding environment
 - ▶ RMarkdown documents can combine text, R code, LaTeX, and any output.
 - ▶ Write in Markdown or Visual Editor

Git and GitHub

- ▶ Git is a version-control system
 - ▶ Manages changes to your code (similar to Track Changes in Word)
 - ▶ Facilitates collaboration and sharing
 - ▶ Helps to ensure reproducibility
- ▶ GitHub is a website used to store code
 - ▶ It is *not* a place to store large files

Terminology

- ▶ A GitHub *repository* (or *repo*) contains all files and associated history
 - ▶ Public and private repos
 - ▶ Files should be organized into directories
 - ▶ GitHub can render Markdown files (suffix `.md`), useful for documentation
- ▶ GitHub repositories exist online and you can *clone* them to your local computer using `git`

Using Git and GitHub

- ▶ Multiple approaches:
 - ▶ RStudio integration (recommended)
 - ▶ Command line (recommended for advanced users)
 - ▶ GitHub Desktop (redundant if using RStudio)
 - ▶ Via web browser (only recommended for viewing)

Using GitHub

Follow the instructions on the course website to set up GitHub with RStudio: https://github.com/t-davidson/computational-sociology/blob/main/2026/course-materials/github_setup.md

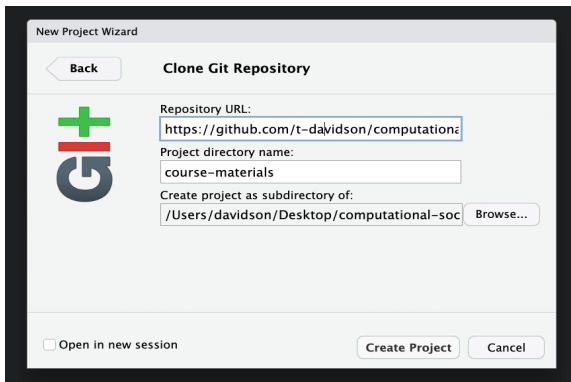
1. Register for a GitHub account
2. Install Git
3. Sync Git with RStudio

Cloning the course materials

- ▶ Once you have this set up, navigate to the course website on GitHub and copy the URL:
 - ▶ <https://github.com/t-davidson/computational-sociology>

Cloning the course materials

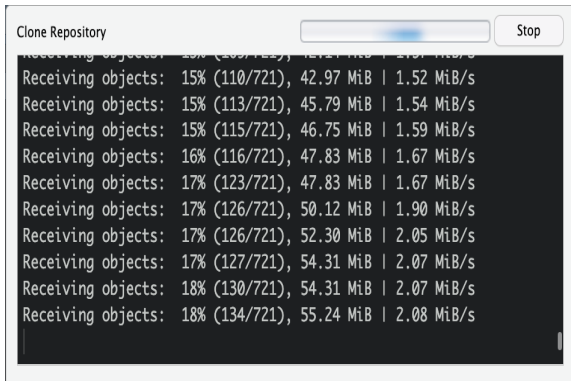
- In RStudio, click File > New Project > Version Control > Git



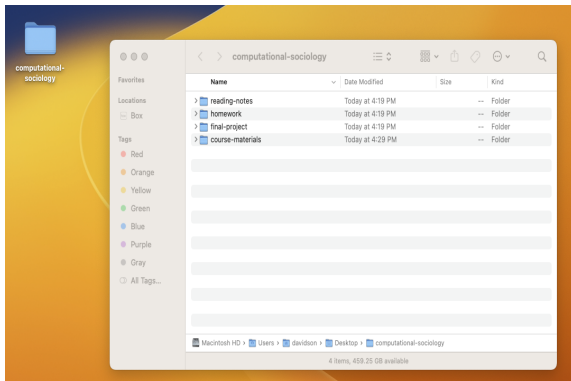
Cloning the course materials

- ▶ Paste the URL into the Repository URL field
- ▶ Write a suitable name in the Project directory name field
 - ▶ This will be the name of the folder that is created on your computer
- ▶ Choose a location to store the repository on your computer
 - ▶ Recommend using the folder we created earlier
- ▶ Then click Create Project

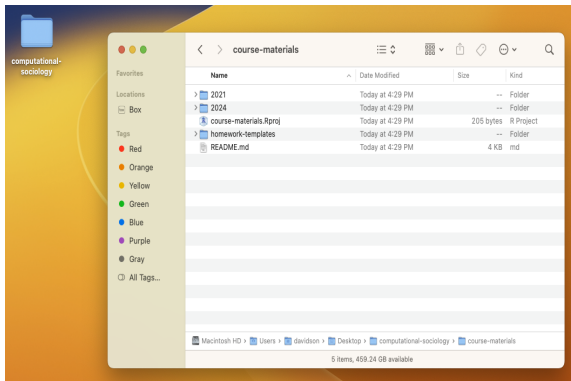
Cloning the course materials



Cloning the course materials

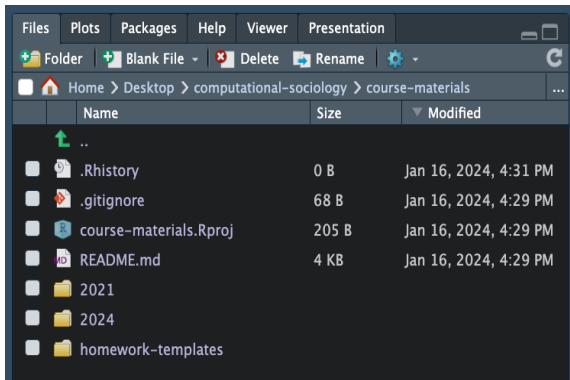


Cloning the course materials



Navigating files using RStudio

You can also use the Files pane in RStudio to navigate folders.



Opening this .Rmd file

- ▶ Navigate to `2026/course-materials/slides/` and open `lecture1-introduction.Rmd` by double-clicking the file

Navigating files using RStudio

Run the `getwd()` command to show the current working directory in R.

```
getwd()
```

Navigating files using RStudio

Run the `setwd()` command to change your working directory. Try going one step back from the current directory using `..`. You can then run `getwd()` to verify it has changed.

```
setwd("..")  
getwd()
```


Navigating files using RStudio

- ▶ When a `.Rmd` file is opened RStudio defaults to the directory where the file is contained.
 - ▶ If you run `setwd()` it will change within a chunk, but other chunks will revert back to the working directory.

Navigating files using RStudio

The working directory is important when considering loading files. Navigate to the code directory. You can use the `list.files` function to see a list of the files in the current directory.

```
setwd("../code/")  
list.files()
```

Using file paths

- ▶ The working directory is critical because it defines a path we need to use to load files. Different files will fall into one of three groups:
 1. Files contained in the working directory.
 2. Files contained in outer directories.
 3. Files contained in inner, nested directories.

Files contained in the working directory

If a file exists in the working directory, you will see it when running `list.files`. It can be loaded by using the file name.

```
library("tidyverse")  
read_file("file.txt") %>%  
  print()
```

Files contained in an outer directory

If a file is contained in an upper level directory, you need to use the `..` to escape the current directory. For each step out from the current directory you need to add another `/..` to the file path. In this case, the file is contained one level above the current directory.

```
read_file("../file_outer.txt") %>%  
  print()
```

Files contained in an inner directory

If a file is contained in an inner directory, you need to add the name of the directory to the file path.

```
read_file("nested/file_inner.txt") %>%  
  print()
```

Files contained in an inner directory

If a file is contained in an inner directory, you need to add the name of the directory to the file path.

```
read_file("nested/nested2/file_inner_inner.txt") %>%  
  print()
```

Exercise

Modify the path below to find and print the contents of the hidden file in the 2021 directory.

```
read_file("") %>%  
  print()
```


File navigation

Common errors

- ▶ It is common to get errors if you misspecify a path when trying to load a file. You will get an error like the following:
 - ▶ Error: 'filename' does not exist in the current working directory ('directory').
- ▶ If this happens, check whether the file is in your current working directory.
 - ▶ You can always use your Files tab or your normal file viewer to verify the location.

Next week

- ▶ Substantive topics
 - ▶ From networks to agent-based models
- ▶ Technical topics
 - ▶ Data structures and algorithms in R
 - ▶ Vectors, matrices, lists, loops, and functions